

ADVENTURE · AI PLAY STRATEGY

Adventure AI Play Strategy
A Post-Hoc Analysis

Why does the AI recommend that move?

The AI sometimes rolls when a move card would draw a card right away, and plays a small multiplier while holding a large one. Both choices have computable reasons. This report breaks down the value VELA and LYRA assign to every outcome of each action, then checks those reasons against card values, deck contents, dice distributions and full play-outs. It also lists where the AI's choices differ from exact values.

1 Reading the game and the AI — rules, base rates, how the AI decides, what one card is worth

2 Choice patterns across full game records

3 Eleven case studies — value tables, outcome breakdowns, card trades, play-outs

4 High-scoring games

5 How reliable the recommendations are — exact endgame values, mistakes, search strengths

AIs analyzed
VELA v4.3 · v4.2 · LYRA v2

Evidence
Rule math · value breakdowns · play-outs · exact values

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SUMMARY

Decision principles at a glance

The numbers come from the case studies and the full game records. The principles interpret how the two AIs calculate; they are not rules that hold in every position.

1 **A card in hand is worth points**

VELA v4.2 values a move card in hand at about 40–55 points in the early and middle game. A ×10 is worth about 72.5 points, a ×2 or ×3 14.9 or 22.4. Playing a card trades that value for distance (Section 1.4).

2 **Drawing a card is a trade**

Drawing with a move card swaps one card in hand for a random card from the deck. At 1,488, playing the +8 (45.3 points) draws a card worth 32.8 points on average and moves the token back 9 tiles. With a weak deck, rolling can beat a move card even when the move card would draw (Cases 1–3).

3 **A roll moves the starting point and keeps the hand**

In front of a stop tile (boss tile), every roll lands on the same tile, and from there the cards already in hand can reach jumps and draw tiles. At 534 the +9 cannot draw, but after a roll to 536 the same +9 takes a jump and draws at 566 (Cases 4–6).

4 **Play the cheap card first**

Small multipliers and negative move cards are worth little in hand. At 597 the AI played the ×3 (12.5 points) and kept the ×8 (53.6 points). The ×3 also led to a draw or a reachable draw on 63.9% of its landings, against 30.6% for the ×8 (Case 9).

5 **The deck decides the next link**

With the same tile and hand, the recommendation changes with the cards left in the deck. At 910, if the last deck card is a +5, the −3 beats a roll by 16.3 points; if it is a +2, the two are within 0.1 points (Case 8).

6 **Spend the hand before the game ends**

A move card is worth only about 15–28 points with 0–4 rolls left, because cards cannot be played after the last roll. Endgame order can be checked against exact values: VELA v4.2 matched the optimal action in 98.8% of endgame states, v4.3 Normal in 99.7% and LYRA v2 in 97.4% (Part 5).

How to read a recommendation

① Where the action really lands (after jumps and stop tiles) → ② what it costs (cards, rolls, a bonus roll) and what it gains → ③ whether the remaining hand connects from the landing tile → ④ the quality of the cards left in the deck → ⑤ the number of rolls left. When two actions are within 1 point, treat them as equally good. Switching to the second-best action at randomly chosen decisions cost about 17.7 points per switch on average, so following every recommendation scores higher on average even though the AI sometimes errs (Section 5.5). The case studies break down VELA v4.2 (= v4.3 Fast); v4.3 Normal, High and Max chose the same action in all eleven cases.

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PART 1

Reading the game and the AI

The rules, base probabilities, the AI's method and the value of a single card, as needed to follow the recommendations.

1.1 Actions and resources

On each turn the player either rolls the dice or plays a card from the hand. Cards come in three types.

Type	Effect	Cost
 Move card	Moves a fixed number of tiles (+1 to +12, −1 to −3). Jumps and draws on the landing tile apply as usual. Does not stop at stop tiles.	One card. Uses no roll (a free action).
 Multiplier card	Moves the sum of two dice times the multiplier (×2, ×3, ×5, ×7, ×8, ×10). Passes over stop tiles.	One card and one roll (the bonus roll, if there is one).
 Stage card	Moves to the first tile of the next stage. The distance depends on the current tile (about 20 tiles on average).	One card. A free action.
Roll	Moves the sum of two dice. Stops at the first stop tile ahead.	One roll (or the bonus roll).

A game starts with 100 rolls, and the number of the final tile is the score. A double on a paid roll (a plain roll or a multiplier card) makes the next roll free; this report calls it the **bonus roll**. A double on the bonus roll does not earn another one. The game ends when no rolls and no bonus roll are left, even if cards remain in hand.

The hand holds at most five cards. With five cards (a **full hand**), landing on a draw tile gives nothing. A played card leaves the hand first, so a move card played from a full hand can still draw on its landing tile. Cards come at random from the **deck** (the cards not yet drawn). The deck starts with 30 cards and is **refilled** to 30 as soon as its last card is drawn.

Board tiles

Draw tile

Landing here draws one card from the deck (nothing with a full hand). 401 of the 2,898 tiles.

Jump tile

Landing here sends the token along a ladder or pipe, forward (up to +34) or back (up to −20). 238 tiles; the destination may be a draw tile.

Stop tile (boss tile)

A plain roll cannot pass it and stops there. 279 tiles; 232 of them form four runs of 58.

1.2 Base probabilities

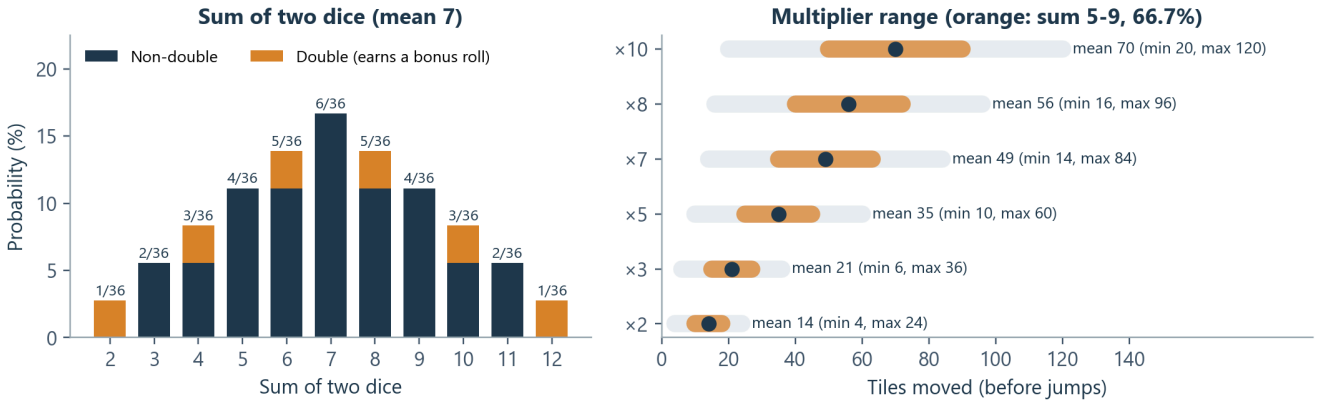


Figure 1: Sum of two dice (left) and the range of each multiplier (right). A multiplier scales the sum, so it stretches the same distribution.

Averaged over the whole board, a roll moves 7.01 tiles and lands on a draw tile with probability 13.1%. Move cards land on a draw tile with similar probability whatever their number (13.7% to 15.0%). A stage card's destination (the first tile of a stage) is never a draw tile. A roll is cut short by a stop tile 19.2% of the time.

Read the probabilities in the case studies against this figure of about 14%. The AI's choice depends on how much the current tile shifts that probability and on whether the hand left after landing leads to the next draw. In the game records, 94.2% (VELA v4.2) and 94.4% (LYRA v2) of the move cards played led to a draw.

1.3 How the AI chooses

Both AIs work the same way. For each action they expand every possible outcome by the rules (dice results, cards drawn), estimate the final score reachable from each outcome, and take the probability-weighted average. That average is the **AI value** of the action (an estimate of the expected final score), and the AI recommends the action with the highest value.

AI	How it values an outcome
VELA v4.2	A large precomputed value table (by tile, rolls left and hand) with a correction for the remaining deck. It looks two steps ahead: the action, then the best reply.
VELA v4.3	The same value table, with a choice of search strength. Fast is identical to v4.2. Normal, High and Max look three steps ahead at every action; when the top two actions are within 1.5 points, they extend the actions within 1.5 points of the best up to six steps. The three differ only in their computation cap (2,400, 47,400 and 383,400 evaluated states). Normal is the service default.
LYRA v2	A neural network (a model trained on many games to predict how good a position is) values each outcome directly. It looks one step ahead.

Over 20,000 games, the search strengths averaged 1,877.2 (Fast), 1,878.6 (Normal), 1,879.3 (High) and 1,879.8 points (Max). Searches of three or more steps do not split cleanly into per-outcome values, so the value tables and outcome breakdowns in the case studies show the two-step calculation of VELA v4.2 (= v4.3 Fast). Each case also lists how v4.3 Normal, High and Max evaluated the same position. In all eleven cases, all four strengths chose the same action as v4.2.

Splitting an action's value by outcome (an **outcome breakdown**) shows which outcomes drove the choice. In every case, recombining the outcome values with their probabilities reproduced the AI value to within 0.001 points.

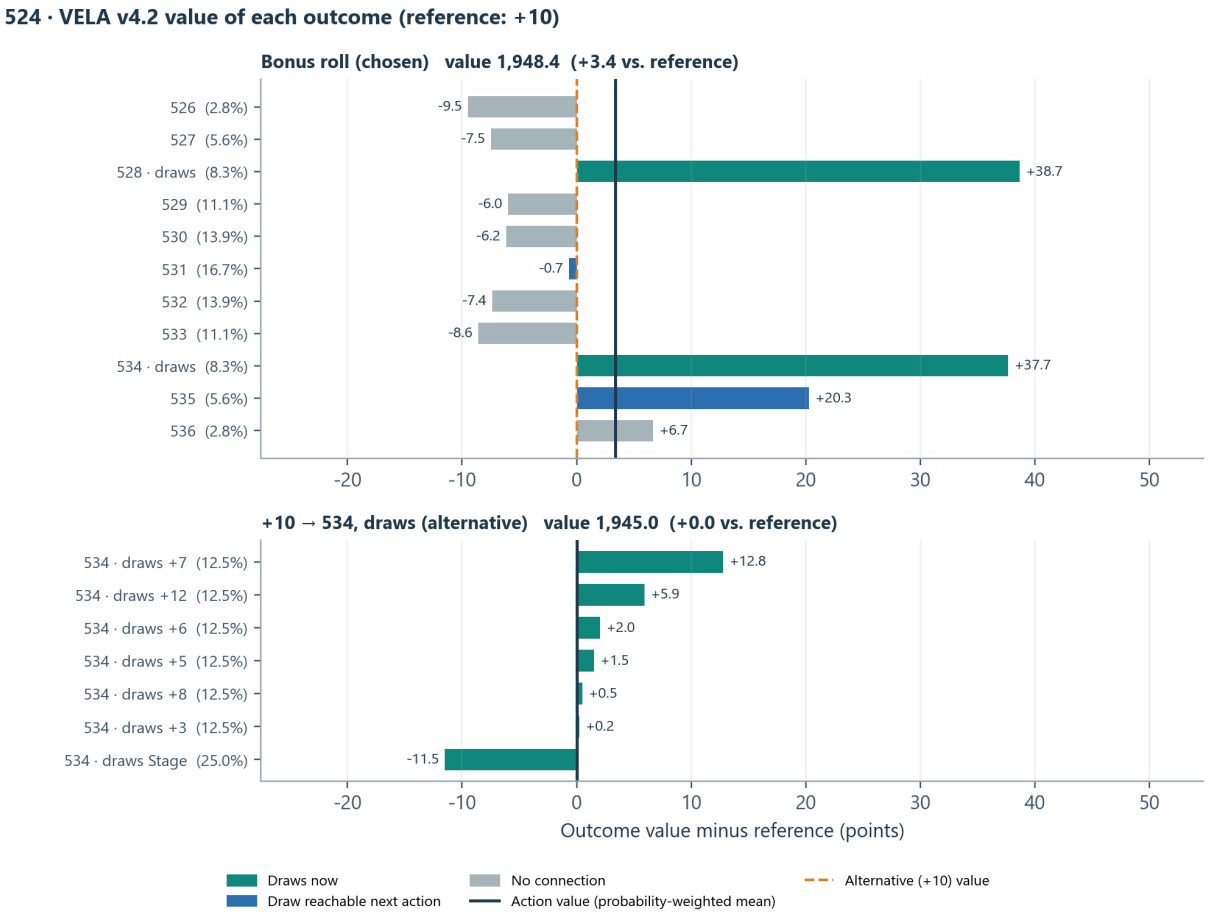


Figure 2: How to read an outcome breakdown (Case 2). Each bar is the AI's value for one outcome minus the reference (dashed orange line, the alternative's value). The solid dark line is the action's own value. Color shows whether the outcome draws a card now (green), leaves a hand that can draw on the next action (blue), or neither (gray).

The explanations rest on four kinds of evidence, each with its own level of certainty.

Evidence	Meaning and limits
Rule math	Moves, jumps, draws and dice probabilities computed from the rules. Exact.
AI value	Action values and outcome breakdowns. They show what the AI based its choice on, but they are estimates, not exact expected scores. Against exact endgame values, the value of the chosen action was off by 0.74 points on average for VELA v4.2, 0.26 for v4.3 Normal and 0.73 for LYRA v2 (Part 5).
Play-out comparison	From the same position, only the first action differs and the same AI plays the rest of the game; the gap in final score is measured over many games. We report play-outs by VELA v4.2 (2,048 pairs) and by v4.3 Normal (512 pairs), with a correction that removes much of the dice luck (a control variate) and a 95% interval (the range likely to contain the true mean gap).
Exact value	The expected final score computed over every remaining possibility. Only feasible in the endgame, with 0–2 rolls left.

1.4 What one card is worth

Playing a card means comparing what the card is worth in hand with what playing it gains now. This report defines a card's **holding value** as the difference in AI value between the same position with and without that card. The figures below come from 22,838 cards in 7,000 positions that VELA v4.2 actually reached.

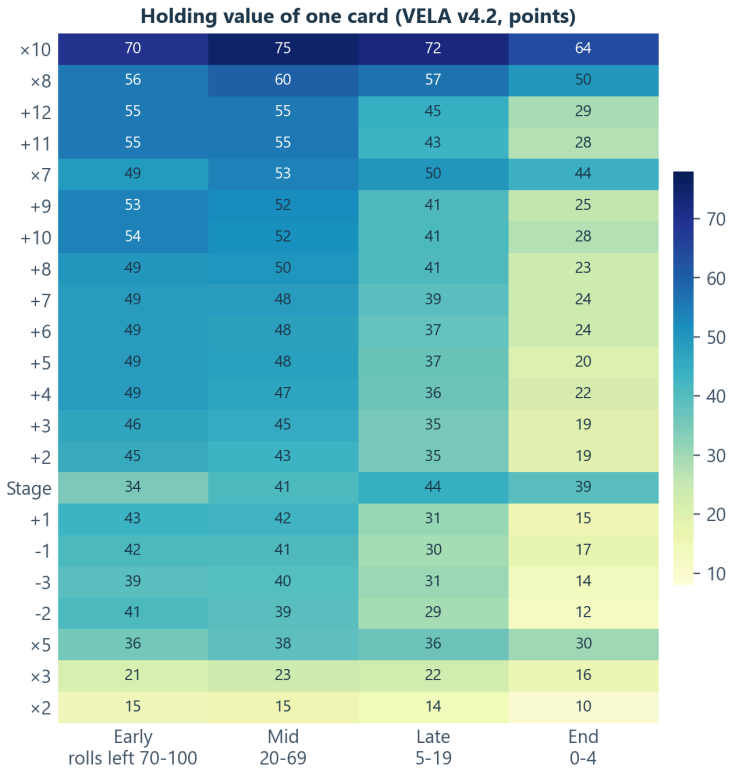


Figure 3: Average holding value by card type and game phase. Phases are defined by the number of rolls left.

The table shows four things.

1. The same move card changes value a lot over the game. A +10 drops from 54.0 points early to 27.7 at the end. With few rolls left there is less time to wait for a good spot to play it, and fewer chances to use the card it draws.
2. Multipliers vary widely. The ×10 is the most valuable at 72.5 points and the ×2 and ×3 the least at 14.9 and 22.4. A ×2 moves only 7 tiles more than a roll on average but costs a card, so the net gain is small.
3. The −1 to −3 cards are worth 37.5 to 38.4 points, about the same as a +1 or +2. Where moving back reaches a jump or a draw tile, a backward card does the same job as a forward one.
4. The stage card rises to 44.2 points late in the game. Stage jumps get longer toward the end of the board (16.6 tiles on average before tile 700, 26.9 after tile 2,500), and a stage card skips a run of stop tiles without using a roll.

Holding value also depends on the tile. At 534 in Case 2, a +11 would reach the +21 jump at 545, so drawing one there would be worth 88.7 points. The case studies read every card play as the following trade.

The trade in a card play

Cost: the holding value of the card played (plus one roll for a multiplier)
Gain: distance, the value of the card drawn if the landing tile is a draw tile, and the next link the remaining hand opens from the landing tile

PART 2

Choice patterns across full game records

How often each AI chose each kind of action in comparable situations, from 10,000 games of VELA v4.2 and 10,000 games of LYRA v2.

2.1 Other actions even when a move card can draw

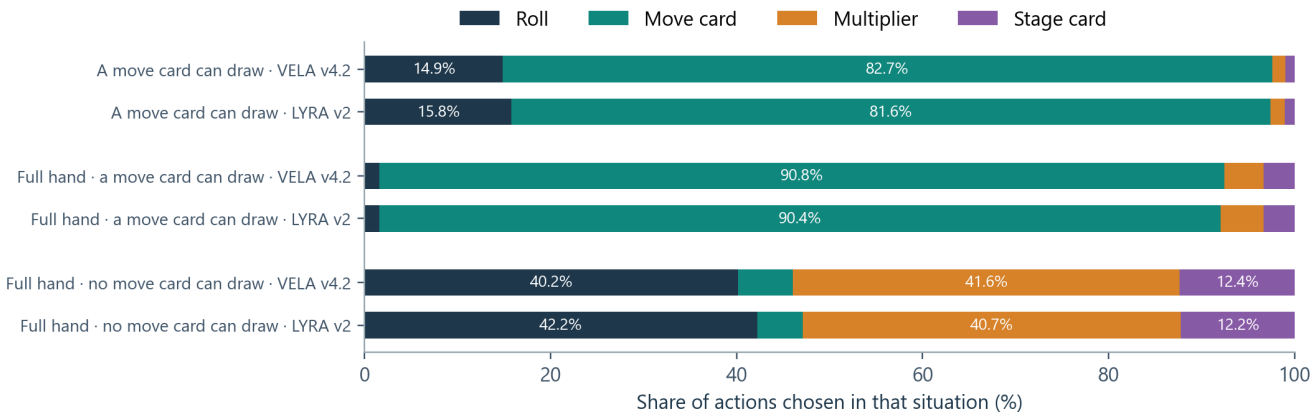


Figure 4: Share of actions by situation. "A move card can draw" means at least one move card in hand lands on a draw tile after jumps.

When a move card could draw (VELA v4.2 522,731 times, LYRA v2 532,490 times), the AIs played a move card 82.7% and 81.6% of the time. In the remaining 17–18% they rolled or played a multiplier or a stage card. With a full hand and no move card that could draw, they mostly rolled (40.2% and 42.2%) or played a multiplier (41.6% and 40.7%). Rather than playing cards to make room, they often kept the hand and changed position.

The two AIs agree to within about one percentage point. LYRA v2, however, started its training from VELA's choices, so it may have inherited similar tendencies. The two data sets also come from different games, so this is not a position-by-position comparison.

2.2 Actions by game phase

Rolls left	Roll	Move card	Multiplier	Stage card
70–100	67.9% · 68.3%	23.4% · 23.0%	6.7% · 6.7%	2.0% · 2.0%
20–69	61.4% · 61.2%	28.3% · 28.5%	7.7% · 7.7%	2.6% · 2.6%
5–19	57.2% · 57.0%	31.5% · 31.6%	8.5% · 8.6%	2.8% · 2.8%
0–4	34.4% · 34.5%	46.0% · 46.3%	14.0% · 13.6%	5.7% · 5.5%

Each cell shows VELA v4.2 · LYRA v2. The denominator is decisions with at least one card in hand. These are shares among the positions actually reached in each phase, not the same positions compared across phases.

Early in the game about 68% of decisions are rolls; in the last four rolls, move cards rise to 46%. This matches the drop in endgame holding values seen in Section 1.4.

2.3 Choosing among multipliers

In positions where the AI held two or more multipliers and played one (51,845 and 51,427 times), it played the smallest one 54.4% and 54.3% of the time. It chose the multiplier with the highest chance to draw on this roll (ties included) 89.8% and 90.0% of the time, and the one with the longest average move 45.6% and 45.7% of the time. Case 9 works through one of these choices.

PART 3

Case studies

Positions where the AI passed up a draw, moved first to set up a card, linked hand and deck, chose between multipliers, or ordered its last cards. Each case comes with the AI's value table, an outcome breakdown, the trade-off of the card played and play-out comparisons.

Reading the value tables

"AI value" is the probability-weighted average of the action's outcome values; the action actually chosen is shaded green and set in bold. "Draws now" is the probability of drawing on the tile the action lands on. "Draw by next action" adds the cases where one move or stage card left in hand can draw from that tile (rule math). Compare AI values only within the same position.

A. A draw is a trade

Drawing with a move card replaces one card in hand with a random card from the deck. The three cases below show this trade losing and winning. The figure puts on one scale the holding value of the card played (orange line) and the holding value of each card the deck could deliver on the landing tile (bars; the count is shown when the deck holds more than one).

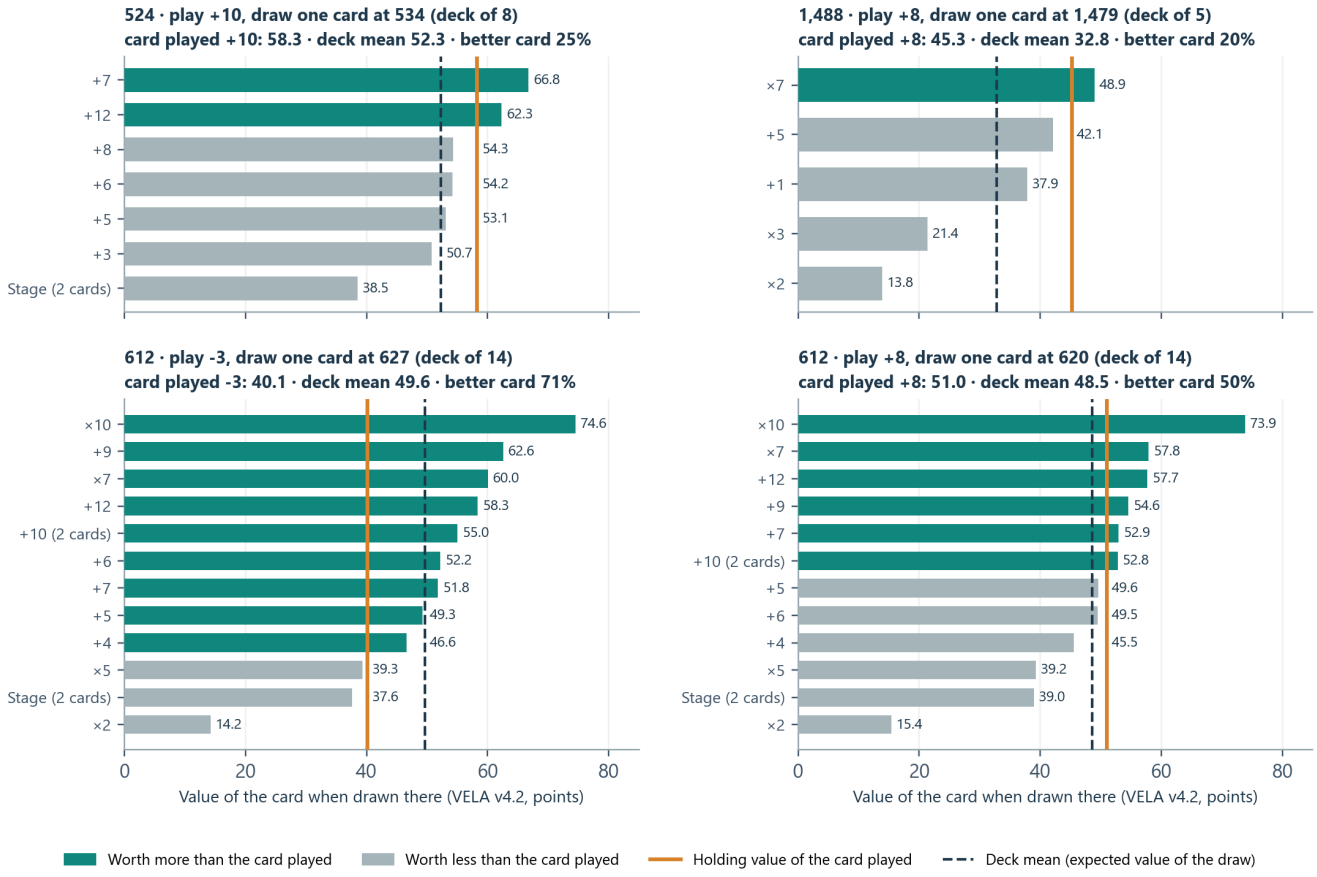


Figure 5: The trade in a draw. Green bars are cards worth more than the card played; the dashed line is the expected value of the draw (deck mean). Values are VELA v4.2 holding values: the difference with and without the card, on the same tile with the same rolls left.

Case 1 · VELA v4.2

A +8 that draws but moves back 9 tiles



Tile **1,488** · **45** rolls left · bonus roll no
Deck: 5 cards (+1 · +5 · ×2 · ×3 · ×7)



Figure 6: The +8 hits the jump at 1,496 (−17) and draws at 1,479, a net 9 tiles back.

Action	AI value	Avg. move	Draws now	Draw by next action
Roll · chosen	**2,424.7**	**+6.4**	**16.7%**	**66.7%**
×10	2,421.5	+73.2	13.9%	44.4%
+8	2,404.9	-9.0	100.0%	100.0%
+7	2,391.8	+7.0	0.0%	0.0%
+10	2,388.9	+10.0	0.0%	0.0%

The +8 is the only card in this hand that can draw, yet VELA v4.2 rolled. It valued the roll 19.8 points above the +8.

What the +8 costs and what it brings

Playing the +8 has three parts.

1. The token moves back 9 tiles.
2. The +8 leaves the hand; at 1,479 its holding value is 45.3 points.
3. One of the five cards in the deck comes in, worth 32.8 points on average. Only the ×7 (48.9 points, 20%) beats the +8, and the ×3 (21.4) or ×2 (13.8), 40% together, lose 24–32 points. A full 30-card deck would average 42.5 points, so this deck is unusually weak.

There is no easy way to win the tiles back. From 1,479 the +10 reaches 1,489 and the +7 then lands on the 1,496 jump again, back to 1,479. Playing the +7 first gives the same loop: 1,486 → +10 → 1,496 → 1,479. The line draws one card, spends two forward cards and ends where it began.

What the AI saw

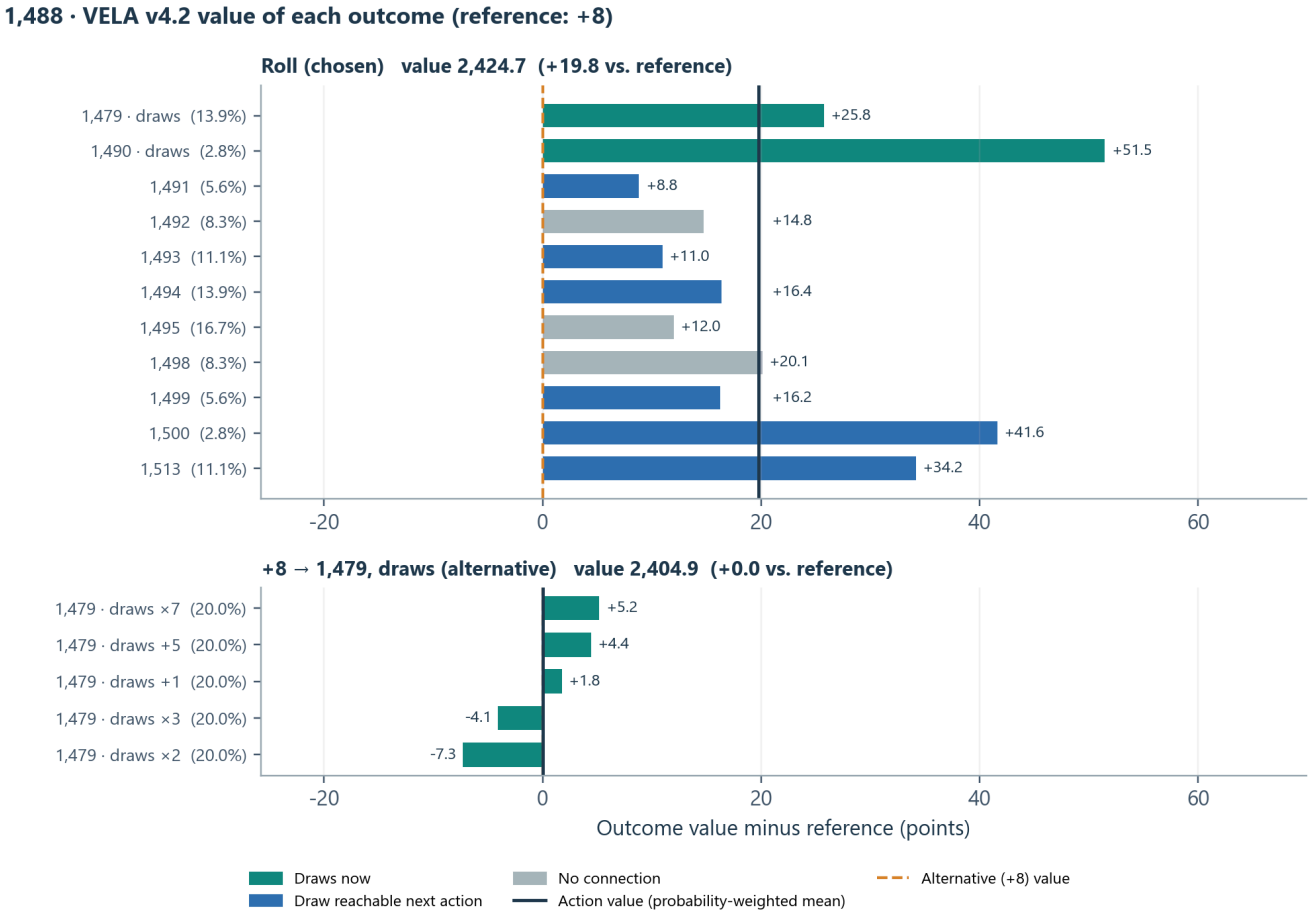


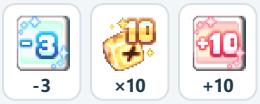
Figure 7: Every outcome of the roll (top) is above the +8's value (dashed orange). The worst is still 8.8 points above.

The roll moves 6.42 tiles on average, draws at 1,479 (via the jump) 16.7% of the time and keeps all four cards. The AI valued every outcome of the roll above the +8. With a full 30-card deck the roll still leads, 2,454.4 to 2,437.1. The weak deck widens the gap, but the main reasons are the 9-tile retreat and the loop.

Evidence	Result
Play-out · v4.2 (roll − +8)	+17.20 points, 95% interval [+15.37, +19.04] · 2,048 pairs
Play-out · v4.3 Normal (roll − +8)	+17.84 points, 95% interval [+14.10, +21.59] · 512 pairs
LYRA v2	Rolls in the same position
VELA v4.2, 1/2/3 steps	Rolls at every depth
VELA v4.3 Normal/High/Max	Rolls at every strength (Normal: 3.02 points above the runner-up)

Case 2 · VELA v4.2

Keep the +10 and take the bonus roll first



Tile **524** · **72** rolls left · bonus roll **yes**
Deck: 8 cards (+3 · +5 · +6 · +7 · +8 · +12 · stage ×2)



Figure 8: The +10 lands on 534, a draw tile. The bonus roll lands on 526–536: it draws at 528 or 534, and from 531 or 535 a card left in hand can draw next.

Action	AI value	Avg. move	Draws now	Draw by next action
Roll · chosen	**1,948.4**	**+7.0**	**16.7%**	**38.9%**
+10	1,945.0	+10.0	100.0%	100.0%
×10	1,925.7	+70.5	0.0%	2.8%
-3	1,899.8	-3.0	0.0%	0.0%

The +10 draws for certain. VELA v4.2 took the bonus roll first instead, by a small margin of 3.4 points. Since the bonus roll costs nothing, the choice is between trading the +10 for a random card now and seeing the roll first. The bonus roll stays available if the +10 goes first.

The eight cards in the deck

Only two cards in the deck are worth more than the +10: the +7 (66.8 points) and the +12 (62.3). At 534 the +10 is worth 58.3 points in hand, and the average card drawn is worth 52.3. The most likely draw is a stage card (25%, 38.5 points). No move card close to the +10 (+9 to +11) and no multiplier is left in the deck. The +10 line beats the roll's value only in the 25% of outcomes that draw the +7 or the +12.

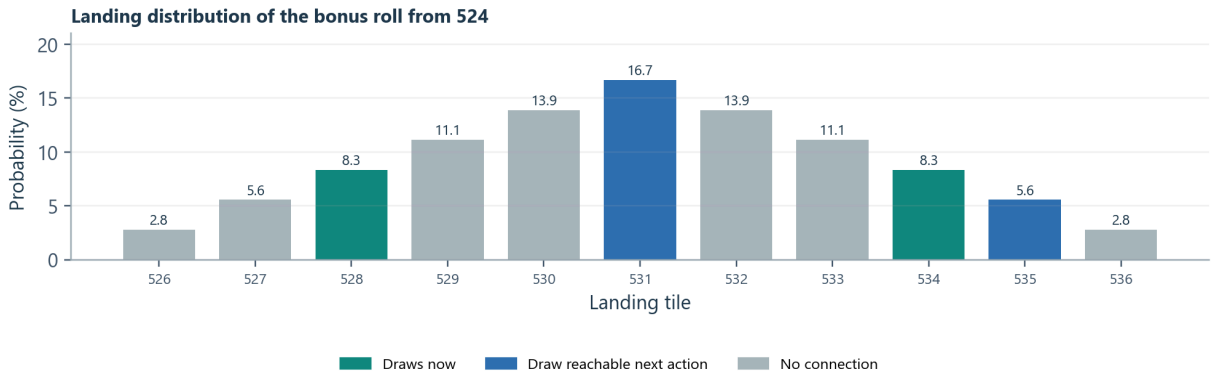


Figure 9: Landing distribution of the bonus roll (rule math). At 528 or 534 (16.7%) it draws and keeps the +10; from 531 (16.7%) the −3 (→ 528) or the +10 (→ 541) can draw; from 535 (5.6%) the +10 can draw via the 545 jump (→ 566).

Risk and reward of the roll

In the example figure of Section 1.3, 25% of the roll's outcomes (528, 534, 535, 536) are 6.7–38.7 points above the +10's value and the other 75% are 0.7–9.5 points below. The good outcomes are large enough to put the weighted average 3.4 points ahead. With a full 30-card deck the gap shrinks from 3.4 to 2.0 points (roll 1,947.7 vs. +10 1,945.7). The weak deck favors the roll more, but the deck alone does not decide the choice.

Evidence	Result
Play-out · v4.2 (roll − +10)	+4.94 points, 95% interval [+2.94, +6.94] · 2,048 pairs
Play-out · v4.3 Normal (roll − +10)	+2.38 points, 95% interval [-1.80, +6.56] · 512 pairs. The v4.2 play-out on the same 512 pairs, +0.47 points, also includes 0 (Section 5.6).
LYRA v2	Rolls in the same position (4.5 points above the +10)
VELA v4.2, 1/2/3 steps	Rolls at every depth
VELA v4.3 Normal/High/Max	Rolls at every strength (Normal: 3.70 points above the runner-up)

Case 3 · VELA v4.2

A −3 that goes further than a +8 and costs less

Tile **612** · **46** rolls left · bonus roll no
Deck: 14 cards



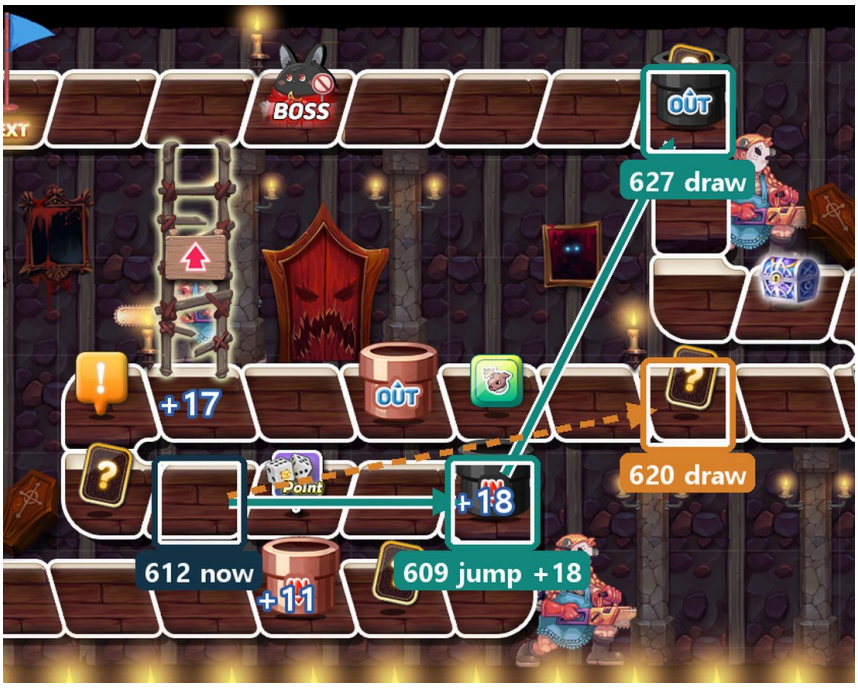


Figure 10: The −3 hits the jump at 609 (+18) and draws at 627. The +8 draws at 620. On this stretch the board runs right to left.

Action	AI value	Avg. move	Draws now	Draw by next action
-3 · chosen	**1,586.9**	**+15.0**	**100.0%**	**100.0%**
+8	1,580.6	+8.0	100.0%	100.0%
×8	1,574.1	+60.3	27.8%	52.8%
Roll	1,570.7	+7.9	13.9%	63.9%
+2	1,524.9	+2.0	0.0%	0.0%

Both the −3 and the +8 can draw. They differ in three ways.

1. The −3 lands 7 tiles further ahead (627 vs. 620).
2. The card played is worth less: on its landing tile the −3 is worth 40.1 points, the +8 51.0.
3. The card kept differs: the −3 keeps the +8, and the +8 keeps the −3.

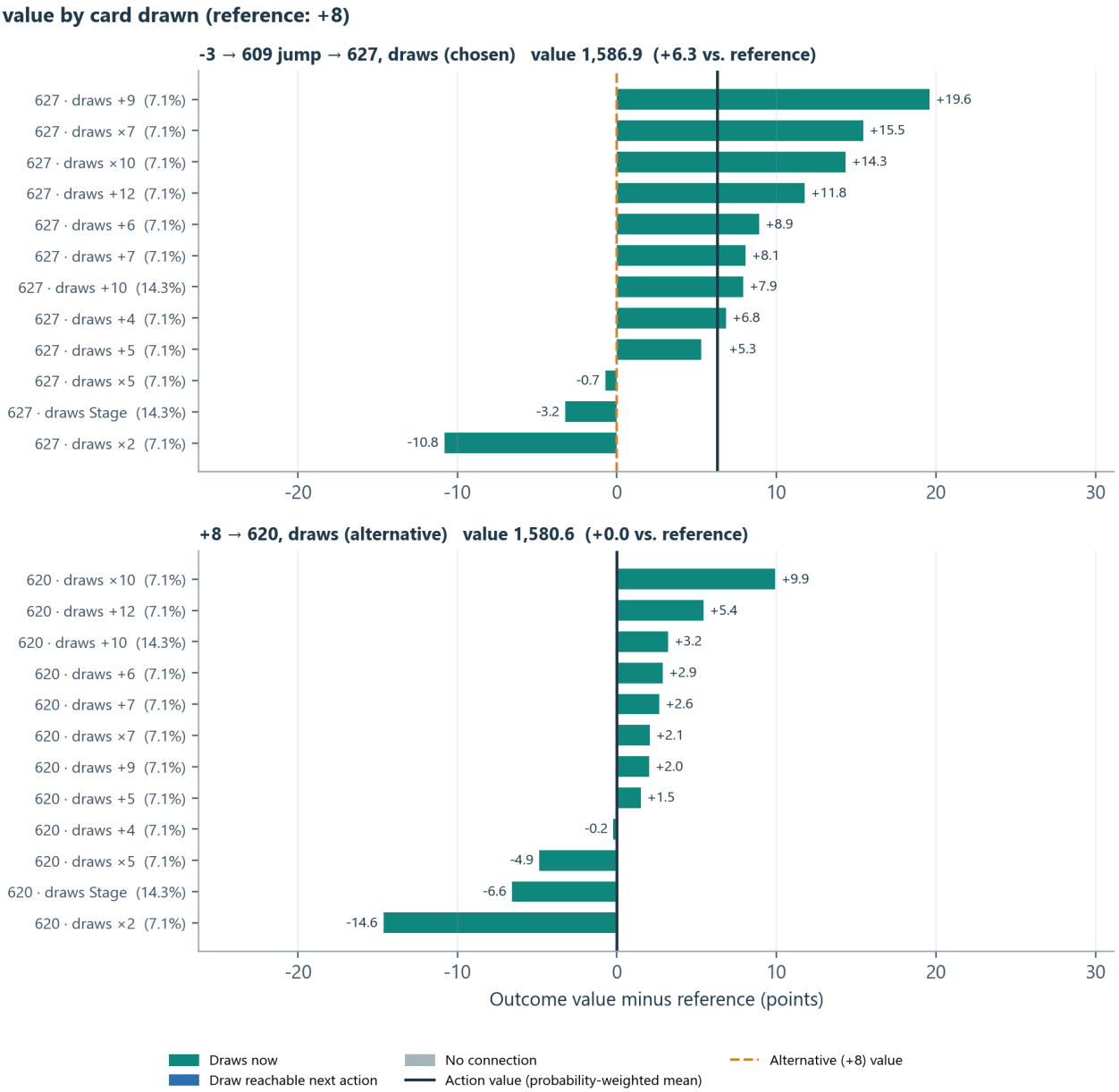


Figure 11: For the same card drawn, the −3 line is valued 3.3–17.6 points higher (the gap is largest when the draw is a +9).

The chance that the deck's 14 cards deliver something worth more than the −3 is 71%, so this trades a cheap card for an average deck card. Despite being a negative card, the −3 here gains ground, draws and keeps the better card. In the game records, 93.6% (VELA v4.2) and 94.0% (LYRA v2) of the negative move cards played led to a draw.

Evidence	Result
Play-out · v4.2 (−3 − +8)	+4.37 points, 95% interval [+2.84, +5.89] · 2,048 pairs
Play-out · v4.3 Normal (−3 − +8)	+2.32 points, 95% interval [-0.69, +5.32] · 512 pairs
LYRA v2	Plays the −3 in the same position
VELA v4.2, 1/2/3 steps	−3 at every depth
VELA v4.3 Normal/High/Max	−3 at every strength (Normal: 6.24 points above the runner-up)

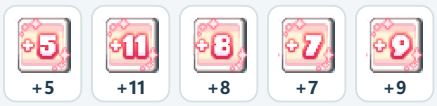
B. Move first, then play the card

Right before a stop tile, a roll always lands on the same tile. A roll then moves the starting point while keeping the hand, so a card that cannot draw now may reach a jump or a draw tile on the next action. In the three cases below the AI rated such a setup move above playing a move card.

Case 4 · VELA v4.2

Roll to the stop tile, then draw twice

Tile **458** · **75** rolls left · bonus roll no
Deck: 8 cards



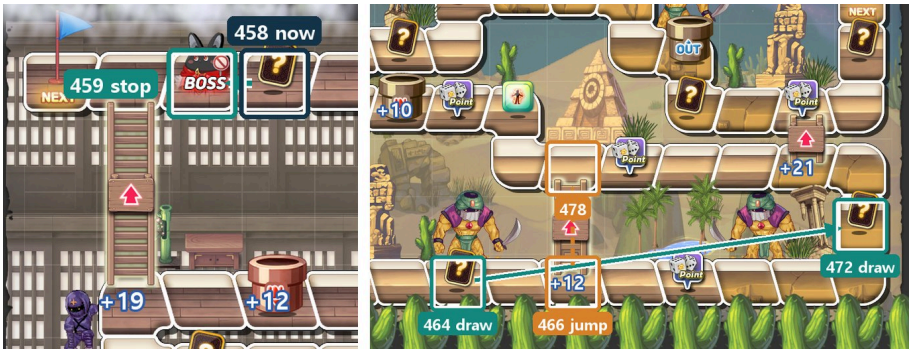


Figure 12: Every roll from 458 stops at 459 (boss tile). From 459, the +5 draws at 464 and the +8 then draws at 472 (green). Playing the +8 now reaches 478 via the 466 jump (+12), without a draw (orange).

Action	AI value	Avg. move	Draws now	Draw by next action
Roll · chosen	**1,974.8**	**+1.0**	**0.0%**	**100.0%**
+8	1,971.3	+20.0	0.0%	100.0%
+7	1,951.6	+7.0	0.0%	0.0%
+5	1,951.2	+5.0	0.0%	100.0%
+11	1,949.6	+11.0	0.0%	0.0%
+9	1,949.5	+9.0	0.0%	100.0%

With a full hand, no card can draw now. A roll lands on 459, and from there the +5 and the +8 already in hand draw twice in a row. Both draws come from the current hand, so the line does not depend on what the deck delivers.

Roll → +5 → +8 ends at 472, playing two cards and drawing two (one roll used). +8 → jump → +7 ends at 485, playing two cards and drawing one (no roll used). The second line is 13 tiles ahead and saves a roll but leaves one card fewer. The roll's lead is only 3.5 points, and much of it comes from doubles: reaching 459 without a double (83.3%) is worth 1,972.2 points, just 0.9 above the +8 (1,971.3), while the doubles that earn a bonus roll (16.7%) are worth 1,987.8. A full hand does not always mean a card should be played first to make room.

Evidence	Result
Play-out · v4.2 (roll − +8)	+4.52 points, 95% interval [+2.46, +6.58] · 2,048 pairs. An earlier comparison on other seeds (1,024 pairs) gave +2.07 points, [−0.98, +5.11], which includes 0.
Play-out · v4.3 Normal (roll − +8)	+6.06 points, 95% interval [+2.02, +10.10] · 512 pairs
LYRA v2	Rolls in the same position (4.6 points above the +8)
VELA v4.2, 1/2/3 steps	Rolls at every depth
VELA v4.3 Normal/High/Max	Rolls at every strength (Normal: 4.74 points above the runner-up)

Case 5 · LYRA v2

Move two tiles to change what the +9 does



Tile **534** · **72** rolls left · bonus roll no
Deck: 14 cards



Figure 13: Every roll from 534 stops at 536 (boss tile). From 536 the +9 hits the jump at 545 (+21) and draws at 566. Played now, the +9 stops at 543 without a draw (orange).

Action	AI value	Avg. move	Draws now	Draw by next action
Roll · chosen	**2,036.7**	**+2.0**	**0.0%**	**100.0%**
×5	2,023.3	+35.8	11.1%	47.2%
Stage	2,008.0	+4.0	0.0%	0.0%
-3	2,004.3	-3.0	0.0%	0.0%
+4	2,000.5	+4.0	0.0%	0.0%
+9	1,989.8	+9.0	0.0%	0.0%

LYRA v2 rolled. Re-evaluated at 536, the +9 is the best action at 2,034.9 points, 18.3 points ahead of a roll (2,016.6). The same +9 moves 9 tiles from 534 but 30 tiles and a draw from 536. It is the most valuable card in this hand (56.3 points), too valuable to spend at 534.

There is also a route without a roll: +4 → 538, stage card → 568, +9 → 577 and a draw. It spends three cards for one, leaving three in hand. The roll route uses one roll, keeps all five cards and earns a bonus roll 16.7% of the time.

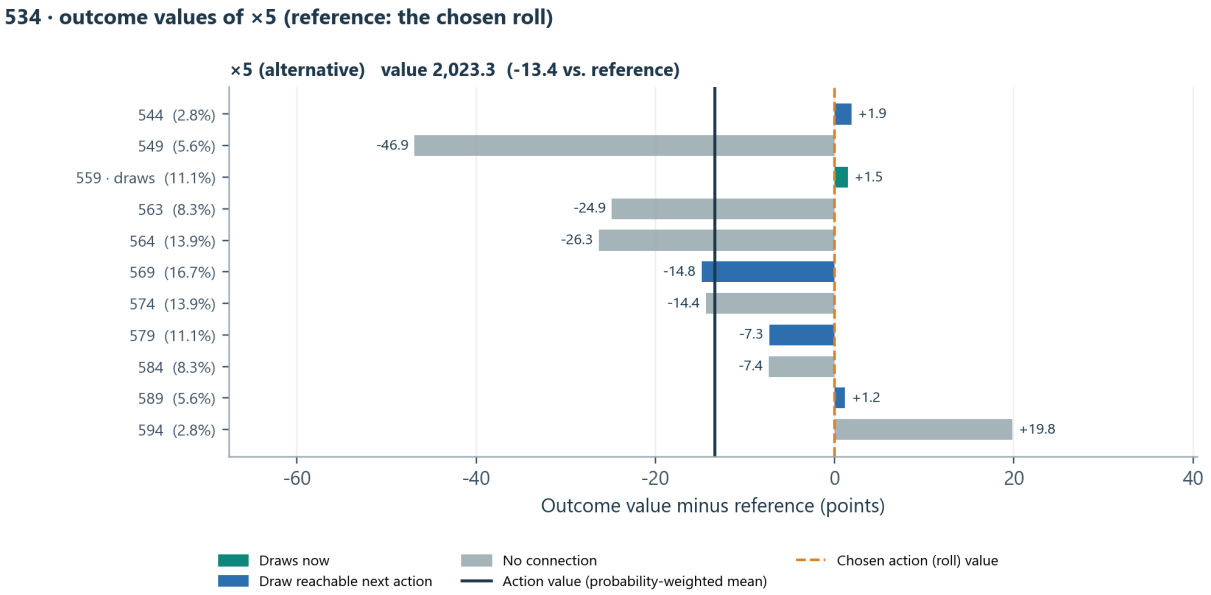


Figure 14: Outcome values of the runner-up, the ×5. Its landings spread widely, and only 25% of them beat the roll's value (dashed orange). The roll has a single outcome, since every roll stops at 536.

Evidence	Result
Play-out · v4.2 (roll − ×5)	+13.44 points, 95% interval [+11.44, +15.43]
Play-out · v4.2 (roll − +9)	+49.36 points, 95% interval [+47.40, +51.33]
Play-out · v4.3 Normal (roll − ×5)	+15.06 points, 95% interval [+11.05, +19.07] · 512 pairs
Play-out · v4.3 Normal (roll − +9)	+50.22 points, 95% interval [+46.13, +54.30] · 512 pairs
VELA v4.2	Same line: roll at 534, +9 at 536
VELA v4.3 Normal/High/Max	Roll at 534 (Normal: 14.48 points above the ×5), +9 at 536

Case 6 · LYRA v2

A stage card moves the +7's starting point

Tile **500** · **70** rolls left · bonus roll no
Deck: 14 cards



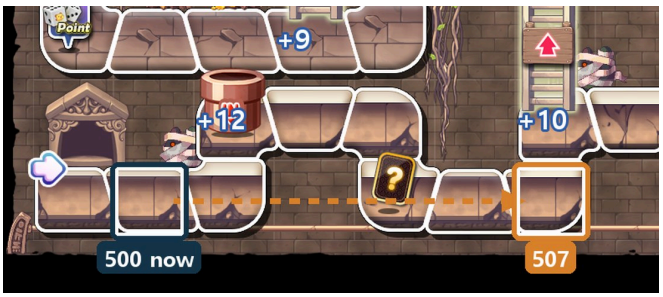


Figure 15: From 500, the +7 stops at 507.

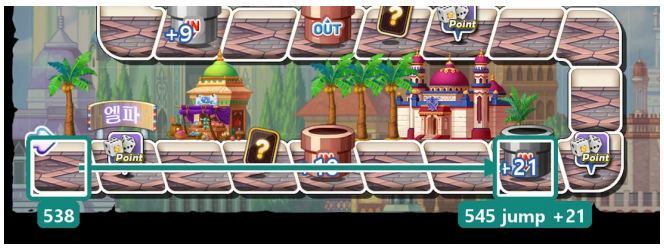


Figure 16: After a stage card to 538, the +7 reaches the jump at 545 (+21).

Action	AI value	Avg. move	Draws now	Draw by next action
Stage · chosen	**1,999.4**	**+38.0**	**0.0%**	**100.0%**
Roll	1,984.8	+8.7	0.0%	72.2%
-1	1,957.7	-1.0	0.0%	100.0%
+1	1,954.7	+1.0	0.0%	100.0%
+7	1,949.9	+7.0	0.0%	100.0%
+9	1,948.1	+9.0	0.0%	100.0%

The stage card to 538 draws nothing. LYRA v2 chose it for the position it leads to. Re-evaluated at 538, the +7 is worth 2,000.5 points, 18.7 above a roll (1,981.8); at 500 the +7 was 49.5 points below a roll. The two cards move 66 tiles and draw one card, and all 70 rolls remain.

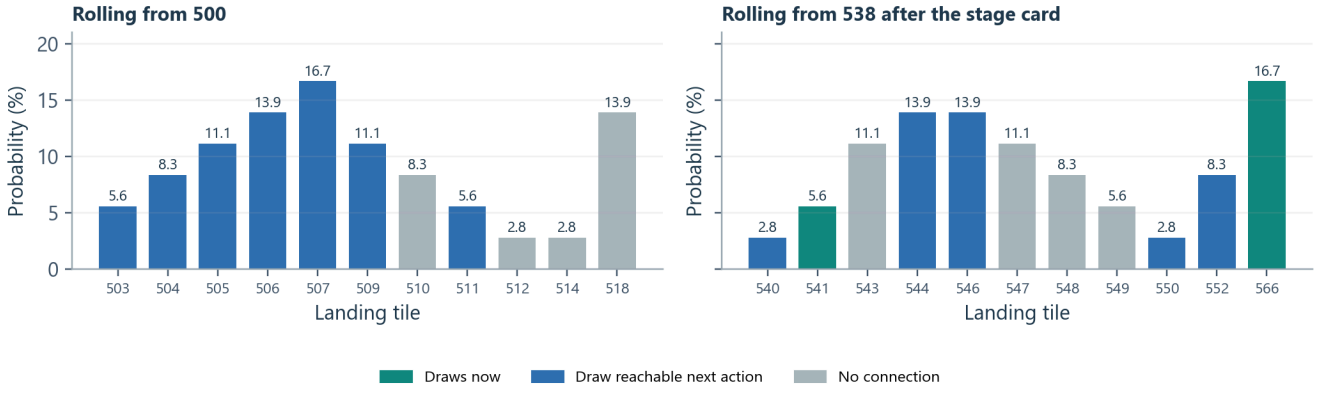


Figure 17: The same roll lands differently and connects differently depending on the starting tile (rule math). From 538, playing the +7 is a sure draw, better than rolling.

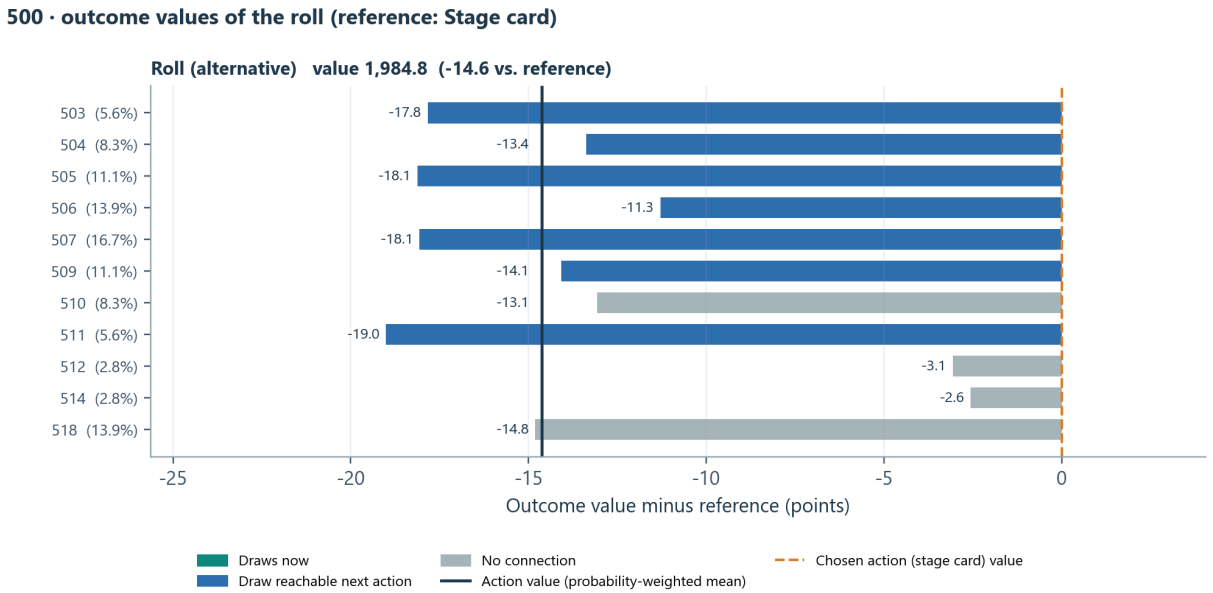


Figure 18: Outcome values of rolling from 500. Only 2.8% of outcomes beat the stage card's value (dashed orange).

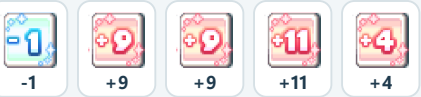
Evidence	Result
Play-out · v4.2 (stage − roll)	+15.75 points, 95% interval [+13.73, +17.77]
Play-out · v4.3 Normal (stage − roll)	+16.28 points, 95% interval [+12.24, +20.33] · 512 pairs
VELA v4.2	Stage card at 500, +7 at 538
VELA v4.3 Normal/High/Max	Stage card at 500 (Normal: 15.48 points above the runner-up), +7 at 538

C. Linking hand and deck

Case 7 · LYRA v2

A first move that opens three draws

Tile **1,033** · **44** rolls left · bonus roll no
Deck: 27 cards



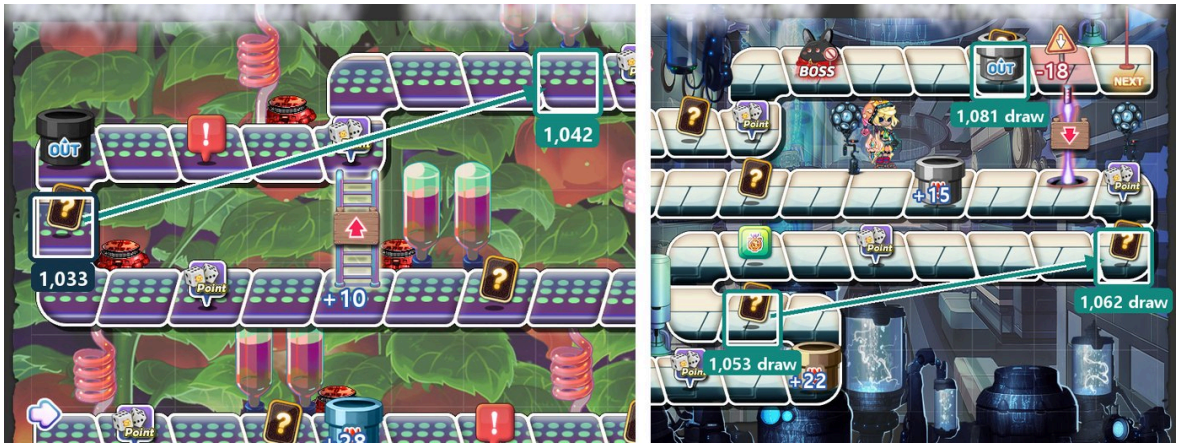


Figure 19: +9 → 1,042, +11 → 1,053 (draw), +9 → 1,062 (draw), +4 → jump → 1,081 (draw). Four cards from the current hand make the whole chain.

Action	AI value	Avg. move	Draws now	Draw by next action
+9 · chosen	**1,978.8**	**+9.0**	**0.0%**	**100.0%**
+11	1,977.0	+11.0	0.0%	100.0%
Roll	1,972.6	+7.0	0.0%	36.1%
-1	1,948.0	-1.0	0.0%	0.0%
+4	1,946.9	+4.0	0.0%	0.0%

No card can draw right now. Playing +9, +11, +9 and +4 in turn, however, draws on the last three landing tiles. Four cards out and three in leaves the hand one card smaller, and the token moves 48 tiles without a roll.

The order of the first two cards does not matter: starting with the +11 (1,044 → +9 → 1,053) builds the same chain. LYRA v2 rates the +9 1.7 points above the +11, and VELA v4.2's two-step search values them equally. Looking only one step ahead, VELA v4.2 would roll instead, because the first move draws nothing and the chain only shows up a step later. VELA v4.3 chose the +9 at every strength; even at Max, searching 5 steps deep, it valued the +9 and +11 equally and 3.6 points above a roll.

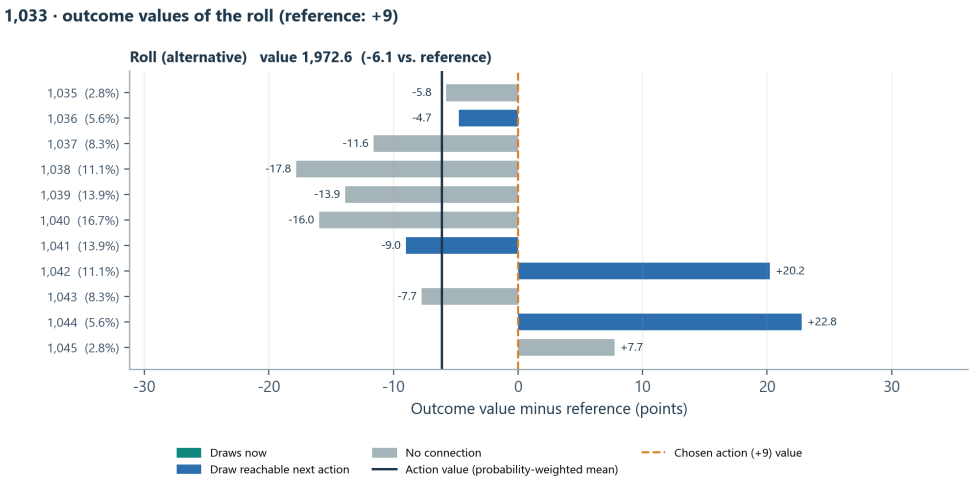


Figure 20: Outcome values of the roll (reference: the +9's value). The +9 has a single outcome (1,042). The roll's outcomes spread out and average 6.1 points lower.

Low confidence

The play-out comparison (+9 − roll) gives +0.79 points, 95% interval [-0.86, +2.43], when VELA v4.2 plays on, and +1.91 points, [-1.29, +5.11], when v4.3 Normal plays on. Both intervals include 0. The three draws are certain under the rules, but these measurements do not show that the chain beats starting with a roll over the full game.

Case 8 · LYRA v2

Is the last deck card a +5 or a +2?

Tile **910** · **43** rolls left · bonus roll no
Deck: 1 card (+5 or +2)





Figure 21: −3 → 907 (draws the last deck card), +9 → 916 (draw), the drawn +5 → 921 jump → 938 (draw). If the last card is a +2, it stops at 918 (orange).

These are two positions with the same tile, rolls and hand, differing only in the last card of the deck. In the records, the AI rolled when that card was a +2 and played the −3 when it was a +5. Drawing the last card refills the deck to 30.

Last deck card +5

Action	AI value	Avg. move	Draws now	Draw by next action
-3 · chosen	**1,778.3**	**-3.0**	**100.0%**	**100.0%**
Roll	1,762.0	+10.2	19.4%	44.4%
Stage	1,734.5	+30.0	0.0%	0.0%
+9	1,724.4	+25.0	0.0%	0.0%

Last deck card +2

Action	AI value	Avg. move	Draws now	Draw by next action
Roll · chosen	**1,754.4**	**+10.2**	**19.4%**	**44.4%**
-3	1,754.4	-3.0	100.0%	100.0%
Stage	1,730.1	+30.0	0.0%	0.0%
+9	1,721.0	+25.0	0.0%	0.0%

With a +5 in the deck, the −3 is certain to draw that +5, and the +5 then reaches the 921 jump from 916. Three cards go out and three come in, so the hand stays at three cards while the token moves 28 tiles. With a +2, the third draw disappears and the −3 and the roll end up less than 0.1 points apart. The +9's holding value shifts with the deck as well: 56.0 points with a +2 and 66.7 with a +5. Whether a follow-up card will arrive at 916 is part of what the +9 is worth.

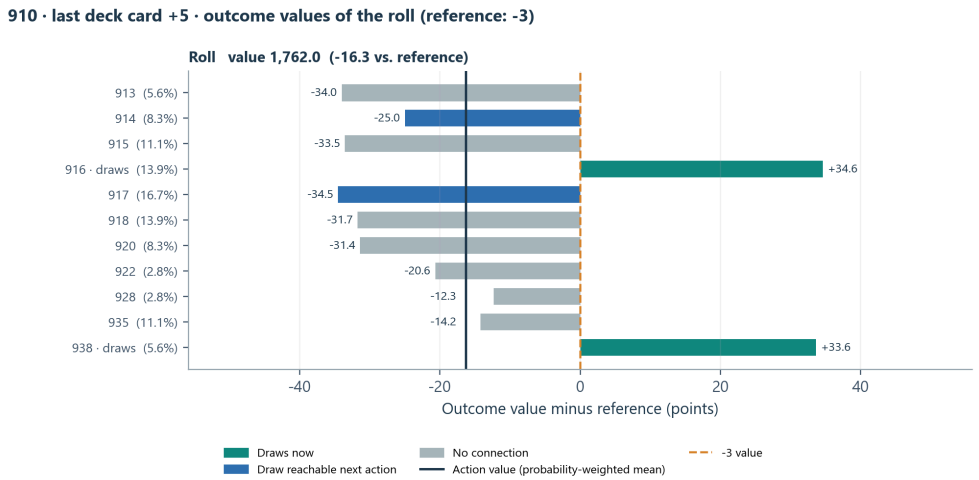


Figure 22: Deck +5: outcome values of the roll (reference: the −3's value). The roll averages 16.3 points below the −3. With a +2 in the deck, the roll's average equals the −3.

Evidence	Result
Play-out · v4.2 (deck +5, −3 − roll)	+15.88 points, 95% interval [+14.37, +17.39]
Play-out · v4.2 (deck +2, roll − −3)	-0.70 points, 95% interval [-2.22, +0.82]. No measurable difference.
Play-out · v4.3 Normal (deck +5, −3 − roll)	+14.30 points, 95% interval [+11.18, +17.43] · 512 pairs
Play-out · v4.3 Normal (deck +2, roll − −3)	-1.08 points, 95% interval [-4.18, +2.01] · 512 pairs
VELA v4.2	Deck +5: −3 (20.3 points above a roll). Deck +2: −3 (0.6 points apart)
VELA v4.3 Normal/High/Max	Deck +5: −3 (17.1 points above a roll) · deck +2: −3 (0.2–0.5 points apart)

With a +2 in the deck the AIs disagree, but by less than 1 point either way. In such close calls either choice costs little (Part 5).

D. Multipliers: landing matters more than distance

Case 9 · VELA v4.2

Play the ×3 before the ×8

Tile **597** · **47** rolls left · bonus roll no
Deck: 14 cards



Action	AI value	Avg. move	Draws now	Draw by next action
×3 · chosen	**1,593.6**	**+25.5**	**16.7%**	**63.9%**
Roll	1,590.1	+8.7	0.0%	38.9%
×8	1,584.5	+56.0	13.9%	30.6%
+8	1,564.7	+8.0	0.0%	100.0%
+2	1,559.4	+2.0	0.0%	100.0%
-3	1,558.3	-3.0	0.0%	0.0%

The ×8 moves 56 tiles on average, the ×3 only 25.5. VELA v4.2 chose the ×3, valuing it 9.1 points above the ×8, for two reasons.

1. The cards are worth different amounts. In this hand the ×3 is worth 12.5 points and the ×8 53.6. Playing the ×3 spends the cheapest card and keeps the most valuable one (the phase averages in Section 1.4 are 22.4 for the ×3 and 58.0 for the ×8).
2. What happens after landing differs. As the figure below shows, 63.9% of the ×3's landings either draw now or leave the +2, +8 or −3 able to draw on the next action, against 30.6% for the ×8. The ×3 tends to land at 612, 618, 627, 630 or 632, close to draw tiles that the remaining move cards can reach.

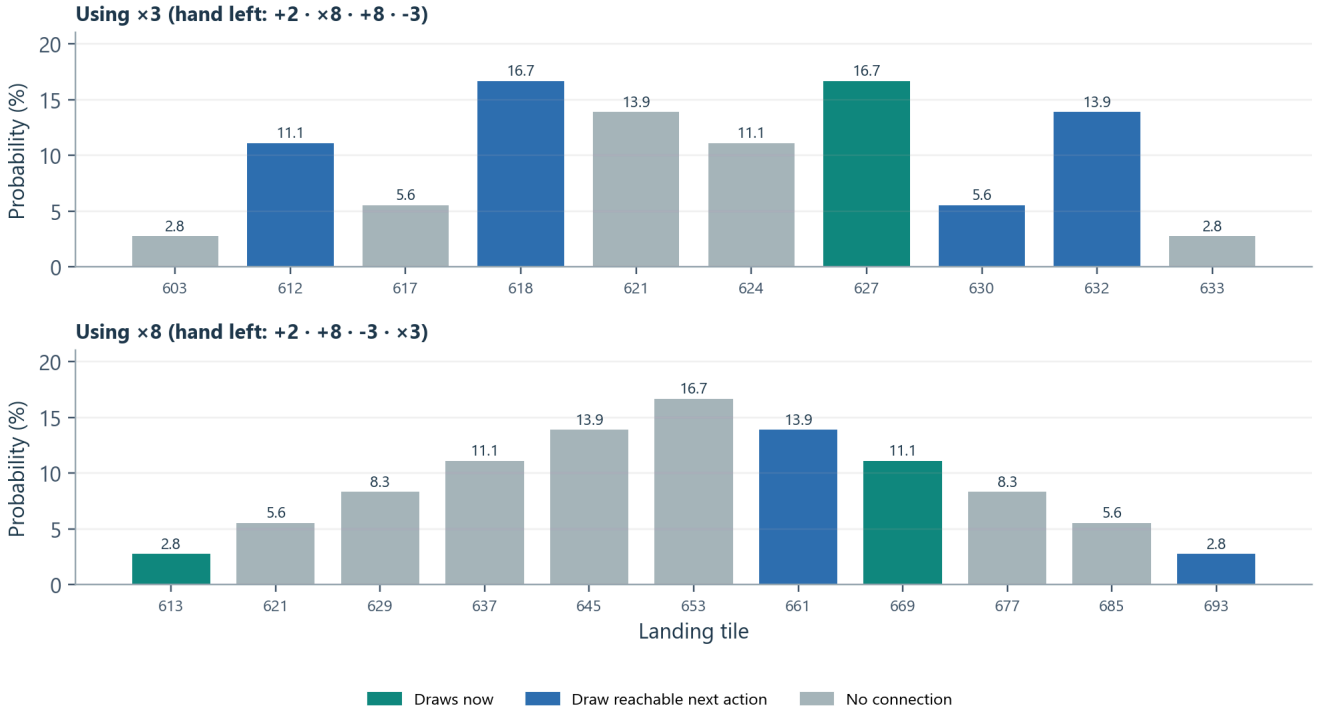


Figure 23: Landing distributions of the ×3 and the ×8 (rule math). Green draws now; blue leaves a hand that can draw on the next action.

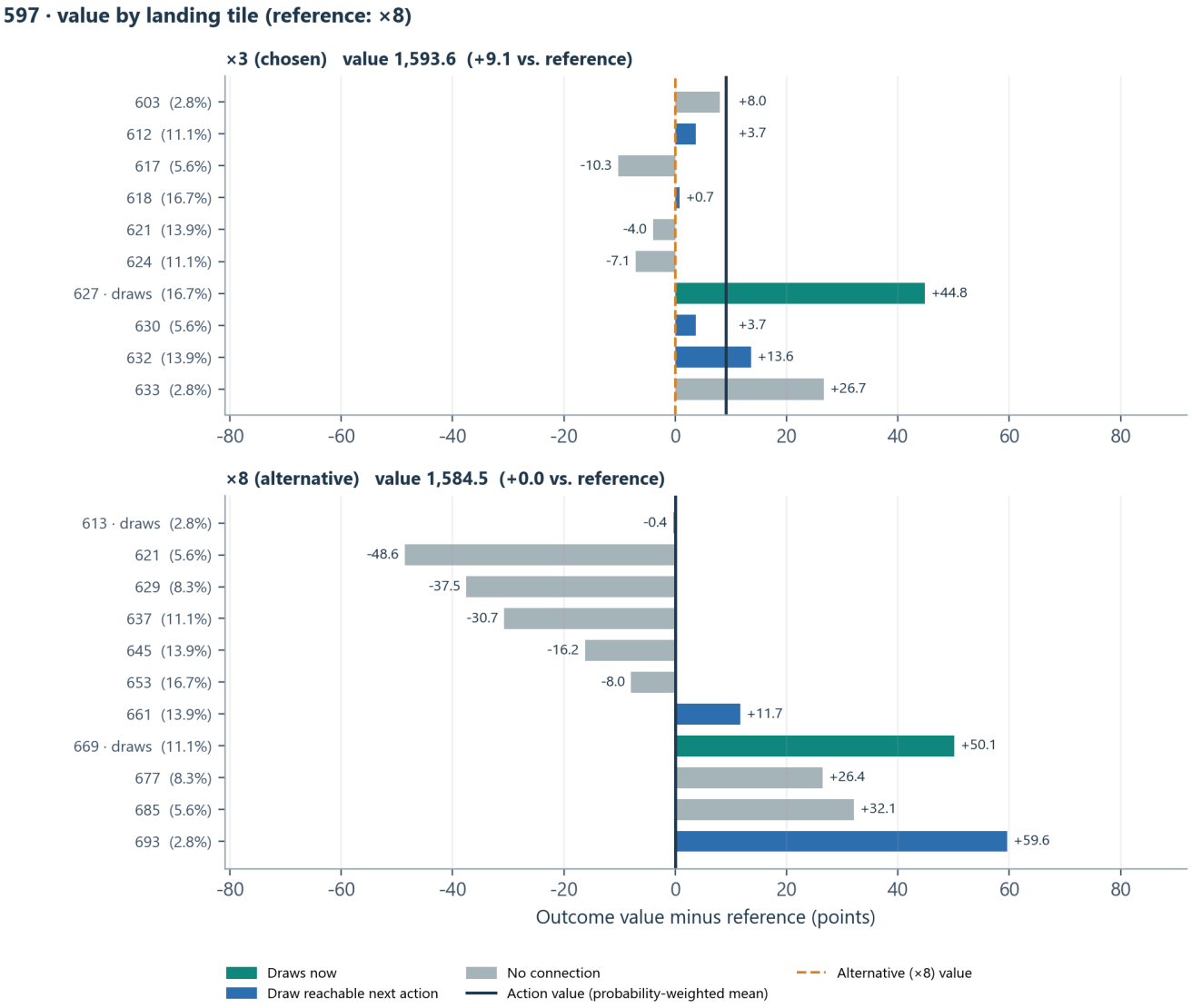


Figure 24: Value by landing tile. 72.2% of the ×3's outcomes beat the ×8's value; only 30.6% of the ×8's outcomes beat the ×3's value.

A different hand changes the choice. Evaluated without the −3, the ×3 falls 10.6 points below a roll, so the preference for the ×3 depends on how this hand links up. The game went 597 → ×3 → 612 → −3 → 609 jump → 627 (draw, Case 3) → ×8.

Evidence	Result
Play-out · v4.2 (×3 − ×8)	+8.32 points, 95% interval [+6.75, +9.89] · 2,048 pairs
Play-out · v4.3 Normal (×3 − ×8)	+8.56 points, 95% interval [+5.49, +11.63] · 512 pairs
LYRA v2	Plays the ×3 in the same position
VELA v4.2, 1/2/3 steps	×3 at every depth; the runner-up is a roll (3.5 points behind)
VELA v4.3 Normal/High/Max	×3 at every strength (Normal: 3.07 points above the runner-up)

E. Endgame: turning the last cards into points

In the endgame the exact value (the expected final score over every remaining possibility) can be computed. Both cases below are explained with exact values.

Case 10 · Top-scoring game (2,813)

Stage cards before multipliers

Tile **2,660** · **3** rolls left · bonus roll no
Deck: —



Tiles 2,660 to 2,717 are a run of 58 stop tiles, so a roll moves one tile at a time. The hand is a ×2, a ×3 and two stage cards.

First action	Exact expected final score	Loss vs. best
Stage card (chosen)	2,815.508	0
Roll	2,814.504	1.004
×2	2,801.500	14.008
×3	2,794.500	21.008

From 2,660 a ×2 lands on 2,664–2,684 and a ×3 on 2,666–2,696; no roll reaches 2,719, the first tile of the next stage. A multiplier played first is wasted, because the stage card that follows goes to 2,719 anyway, and the multiplier and a roll are gone. Playing both stage cards first reaches 2,779 with three rolls, the ×2 and the ×3 still in hand, and every multiplier move after that adds directly to the final tile.

At 2,779 the order barely matters: by exact values the ×2 is best and the ×3 actually played is 0.0006 points lower. Putting the stage cards first was worth 14–21 points; the multiplier order afterward is effectively a tie.

Case 11 · VELA v4.2 · a position from play

A roll at the end of a stop run goes further

Tile **2,677** · **1** roll left · bonus roll no
Deck: —



How far a roll moves depends on the distance from the starting tile to the next stop tile. Inside a run of stop tiles the next tile is always a stop tile, so every roll moves one tile. On 2,717, the last tile of the first run, the next stop tile is 2,720: a sum of 2 moves two tiles (2,719) and any sum from 3 to 12 moves three (2,720). Whether the roll is paid or a bonus roll makes no difference to the distance.

At 2,677 VELA v4.2 played the ×10. A (2,2) or (5,5) lands on 2,717 or 2,777, the last tile of a stop run, and as a double it earns a bonus roll. That bonus roll starts at the end of the run and moves 2.97 tiles on average. Other doubles land on 2,697, 2,737, 2,757 or 2,797, where the bonus roll moves just one tile. The exact expected final score of this position is 2,747.276. Counting every roll as one tile gives a value 0.110 points lower; the difference comes from landings at the end of a run, such as 2,717.

In the actual game the ×10 rolled a sum of 8, and the game ended at 2,757.

PART 4

High-scoring games

The score distribution of one million VELA v4.2 games, and how the games that scored 2,600 or more used their resources.

4.1 One million games

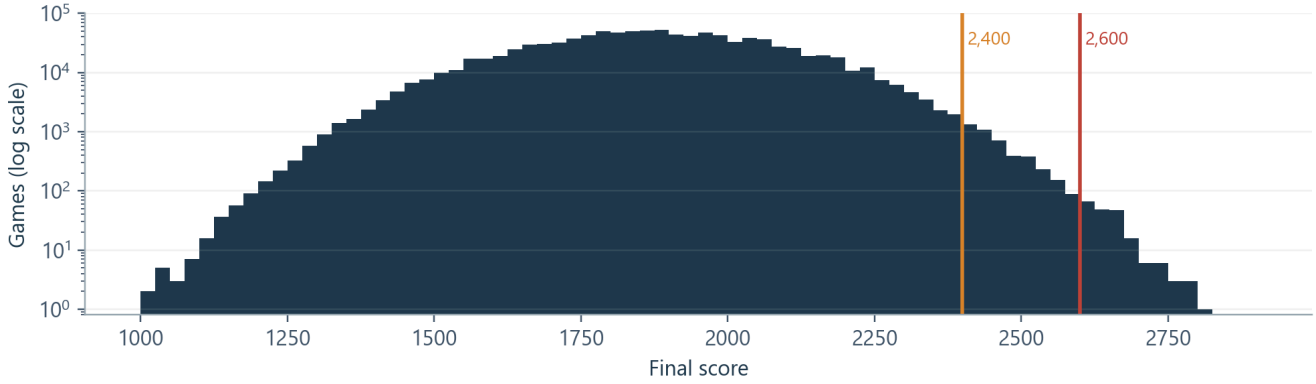


Figure 25: Final scores of one million VELA v4.2 games (log scale).

The mean is 1,877.0 points and the standard deviation (how widely scores spread from game to game) is 201.2. Half of the games end below 1,874, the top 1% reach 2,343 or more and the top 0.1% 2,501 or more. The best game scored 2,813 points, and no game reached the end of the board at 2,898.

Final score	Games	Share
2,000 or more	271,463	27.1463%
2,400 or more	4,558	0.4558%
2,600 or more	196	0.0196%
2,700 or more	19	0.0019%
2,800 or more	1	0.0001%

4.2 Resources in high-scoring games

We replayed all 196 games that scored 2,600 or more and 130 games from lower score bands to collect their action logs. Final scores, action counts and action sequences all matched the original records.

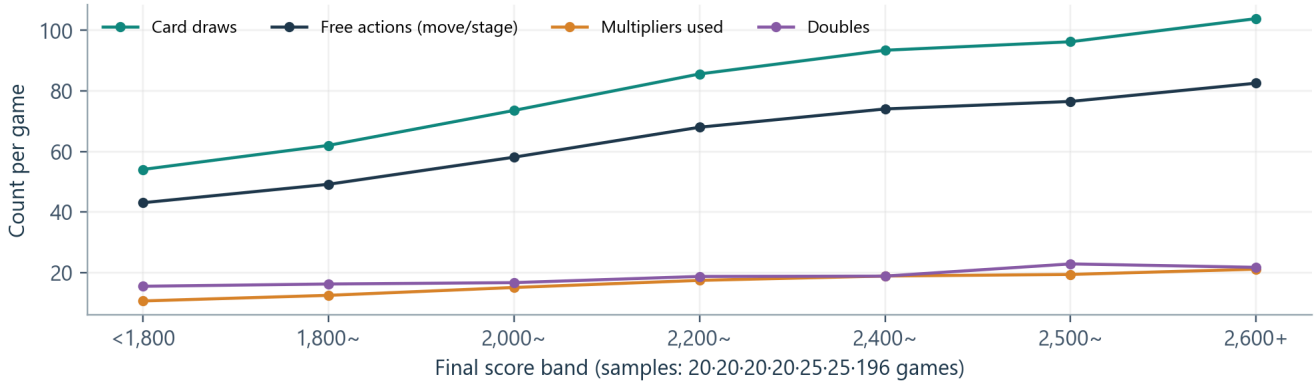


Figure 26: Actions per game by final score band. Higher bands drew more cards, took more free actions and played more multipliers.

Higher-scoring games drew more cards (about 104 per game at 2,600 or more, with about three deck refills), took more free actions and played more multipliers. This comparison groups games by their final score after the fact. Good dice raise both the number of draws and the score, so it does not show that drawing more cards causes higher scores.

What was left on first passing tile 2,000

Final score group	Tile reached	Rolls left	Cards in hand
2,600 or more (196 games)	2,020.0	21.0	4.2
2,400–2,599 (50 games)	2,019.2	16.0	4.0
Below 2,400 (sampled games that reached 2,000)	2,020.3	5.1	2.6

Games that finished at 2,600 or more still had 21 rolls left on average when they first passed tile 2,000. Games at 2,400–2,599 had 16, and lower-scoring games 5. The higher the final score, the earlier the game reached 2,000 and the more rolls it had left. This too is an observed difference, and it includes the good dice luck up to that point.

4.3 The top game: 2,813 points

Over 209 actions this game drew 106 cards (with three deck refills), played 22 multipliers and took 84 free actions. Near the end it crossed the stop run starting at 2,660 with two stage cards (Case 10). By exact values its final multiplier order (×3 first) was 0.0006 points below the best (×2 first). Even the top game did not play every move optimally, and copying its line would not reproduce its score.

PART 5

How reliable the recommendations are

How often the AI is right, how much it loses when it is wrong, and what happens when a player overrides it.

5.1 What reliability means here

Reliability splits into two questions: is this one recommendation the best move, and how many points does following the recommendations score on average? The first can be checked directly only in the endgame, where exact values can be computed. The second is measured by playing thousands of games and averaging the scores.

The decisions in a typical game look like this (decisions with cards in hand, 2,000 VELA v4.2 games).

161 decisions per game

Decisions where the hand offers a choice. About 4 of them come with 0–2 rolls left.

About 11 close calls

Decisions where the AI rates the best and second-best actions less than 1 point apart. Either choice costs little.

Most gaps are large

The second-best action is rated 9.4 points below the best (median). Only 19% of decisions have a gap under 3 points.

Measured against exact endgame values, VELA v4.3 Normal chose a non-optimal action about 3 times per 1,000 decisions and lost more than 1 point 1.4 times. VELA v4.2 did so 12 and 5.1 times, and LYRA v2 26 and 7.4 times.

5.2 Comparison with exact endgame values

A catalog holds the exact value of every action in 4,312 positions reached in real games with 0–2 rolls left. We compared the action each AI chose with the exact optimal action. VELA v4.3 was run with the production search code; at Fast it chose the same action as v4.2 in every position.

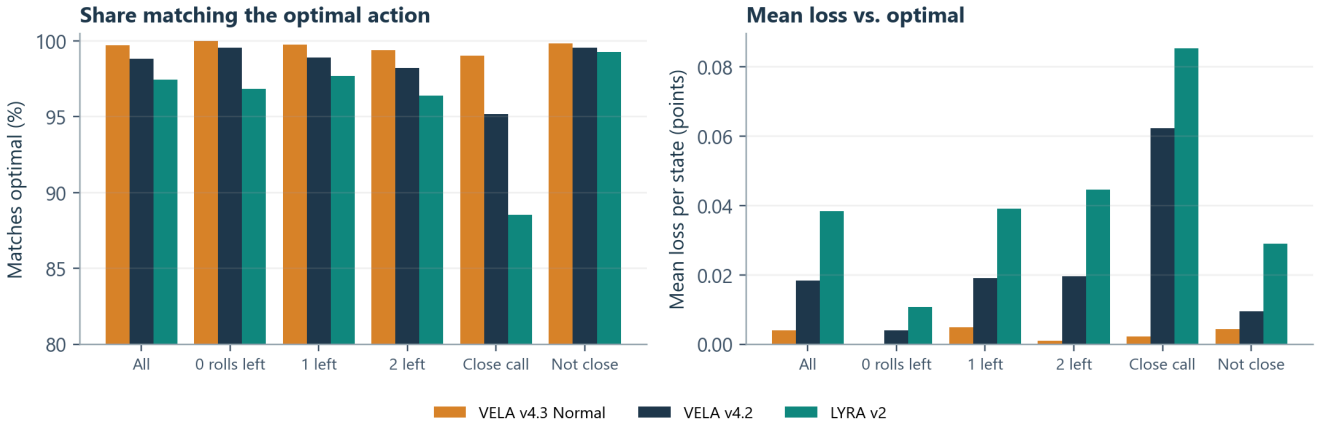


Figure 27: Share of positions where the AI matched the optimal action (left) and mean loss per position (right). The loss is the exact value of the optimal action minus the exact value of the action chosen. A close call is a position where the top two actions are less than 1 point apart in exact value.

Measure	VELA v4.3 Normal	v4.3 High/Max	VELA v4.2	LYRA v2
Matches the optimal action	99.7%	99.8%	98.8%	97.4%
Mean loss per position	0.004	0.003	0.018	0.038
Positions losing more than 1 point	0.14%	0.12%	0.51%	0.74%
Largest loss	4.1	4.1	14.8	14.8
Match rate in close calls	99.0%	99.6%	95.2%	88.5%
Error in the chosen action's value	0.26	0.25	0.74	0.73

Losses and errors are in points.

VELA v4.2 and LYRA v2 chose the optimal action in more than 97% of endgame positions, and 70–75% of their mistakes came in close calls (17% of positions). VELA v4.3 Normal searches deeper in close calls, which raises its close-call match rate from 95.2% to 99.0%. It corrected 44 of the 51 positions where v4.2 went wrong and newly erred in 6 positions where v4.2 was right (losing up to 4.1 points). High and Max gave identical results on this catalog.

The catalog only includes positions that could be solved within a node limit, so it covers only about 22% of positions with two rolls left. In the endgame a six-step search nearly reaches the end of the game. There is no evidence of an equally large gain in the middle game; over whole games, Normal beats Fast by 1.4 points on average across 20,000 games (Part 1). Exact values cannot be computed before the endgame, so these match rates do not carry over to the game as a whole.

5.3 Why LYRA scores higher despite more endgame errors

In the endgame comparison LYRA v2 made more mistakes than VELA v4.2. Yet over the same 9,200 games, with the control-variate correction, LYRA v2 averaged 1,879.95 points and VELA v4.2 1,877.30, a lead of 2.65 points for LYRA. The two results measure different parts of the game. A game has only about 4 decisions with 0–2 rolls left, so the endgame gap of 0.020 points per position amounts to about 0.08 points per game. Nearly all of the 2.65-point lead comes from the remaining 157 or so decisions.

A comparison that mixed the two AIs by phase over the same games shows where the lead comes from.

Phase played by VELA v4.2 (LYRA v2 plays the rest)	vs. LYRA v2 alone
First 20 rolls (80 or more left)	−0.13 (no difference)
First 50 rolls	−1.20
First 80 rolls	−2.12
Whole game	−2.65
Last 20 rolls (fewer than 20 left)	−0.40

Differences in points.

The two AIs are even over the first 20 rolls; the gap opens mostly between 80 and 20 rolls left. Handing only the last 20 rolls to VELA v4.2 still costs −0.40 points, so LYRA v2 is better over the last 20 rolls as a whole, and VELA v4.2 is slightly more accurate only in the very short stretch with 0–2 rolls left. VELA v4.2's value table comes from a compressed exact calculation, which makes it strong in the endgame, where its two-step search reaches the end of the game. LYRA v2 is a neural network trained on its own games, and it appears to capture long-range links in the middle game (position, hand and deck) better than VELA v4.2. This report did not test which factor causes the difference.

VELA v4.3 Normal scores 1.4 points above v4.2 over 20,000 games, which would narrow the gap to LYRA v2 to a little over 1 point (not measured on the same games). Since the game is scored by the final tile, the average score is the measure to use when choosing an AI; the endgame match rate covers only a few decisions per game.

5.4 A case where the AI was wrong

VELA v4.2 and LYRA v2 made the same mistake

Draw before taking the jump

Tile **1,636** · **1** roll left · bonus roll no
Deck: 4 cards (+8 · ×5 · ×8 · stage)



First action	Exact value	Loss vs. best
+7 (chosen by VELA v4.2 and LYRA v2)	1,680.9	−14.8
−1 or −3 (best; chosen by v4.3 Normal and above)	1,695.7	0
+9	1,665.8	−29.9
Roll	1,661.1	−34.6

VELA v4.2 and LYRA v2 played the +7, took the jump at 1,643 and advanced 23 tiles to 1,659. The exact best line is −1 → 1,635, +7 → 1,642, −3 → 1,639 (a draw tile): spend the free cards to draw first, then use the drawn card and the remaining +9 to keep going before the last roll. With one roll left, the order in which the remaining cards are spent decides the score, and a two-step search cannot see that difference. We confirmed this case with an independent exact calculation, which gave the same values to three decimal places.

VELA v4.3 Normal, High and Max searched this position 6 steps deep and chose the −1. Its value for the −1, 1,695.75, was within 0.1 points of the exact 1,695.69. Normal also made new mistakes. The largest is at 1,632 (one roll left, hand +7 · stage · −1 · +7). The three-step search rated the stage card 1.95 points above the +7; the gap exceeded 1.5 points, so it searched no deeper. By exact values the +7 that v4.2 chose is best and the stage card is 4.1 points worse. Deep search only runs in close calls within 1.5 points, so positions where the three-step value is off by more than 1.5 points remain wrong.

5.5 Overriding the AI

We measured what it costs to replace the recommendation with another action only at decisions where it looks wrong. VELA v4.2 played 3,000 games for each override rule, with the same dice sequences across rules, and we compared control-variate-corrected average scores. Following every recommendation averaged 1,877.8 points.

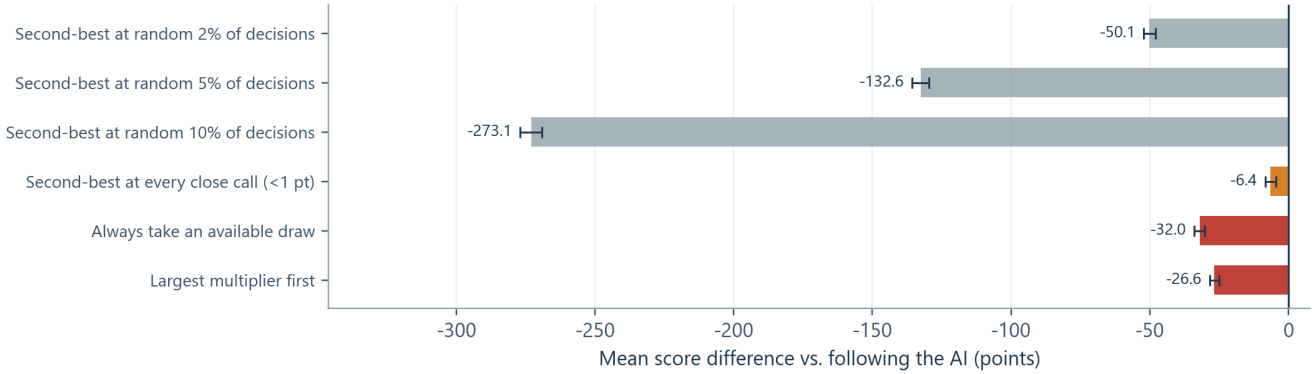


Figure 28: Average score lost under each override rule (95% intervals), compared game by game on the same dice sequences.

Override rule	Change per game	Overrides per game	Loss per override
Second-best at a random 2% of decisions	-50.1	3.1	16.0
Second-best at a random 5% of decisions	-132.6	7.5	17.7
Second-best at a random 10% of decisions	-273.1	13.3	20.6
Second-best at every close call (gap < 1 point)	-6.4	13.0	0.5
Always play a move card that can draw	-32.0	8.0	4.0
Always play the largest multiplier first	-26.6	2.6	10.1

Values in points.

Picking the second-best action only about three times per game lowers the average score by 50.1 points, because at most decisions the second-best action is several points worse than the best. Overriding only close calls costs little, but without the AI's values there is no way to tell which decisions are close. "Never pass up a draw" and "play the big multiplier first" are reasonable instincts, yet, as the case studies show, they often lose depending on the deck, the tile and the remaining hand, and turning them into fixed rules lowers the average score.

The AI does make mistakes, but they are rare and mostly come in close calls, so they cost little (0.004 to 0.038 points per position in the endgame). The decisions a player overrides because they "look wrong" include ones with large gaps; in the random-override experiment each override cost 16.0 to 20.6 points on average. The AIs' average scores (about 1,878.6 for VELA v4.3 Normal and about 1,880 for LYRA v2) assume the recommendation is followed at every decision; overriding some of them does not keep that average. In this experiment VELA v4.2 played the rest of each game, and the decisions a real player overrides may be distributed differently from these rules.

5.6 Evidence for middle-game decisions

Middle-game decisions have no exact values, so they are checked with play-outs and by comparing the AIs. The eleven comparisons from Part 3 are collected below.

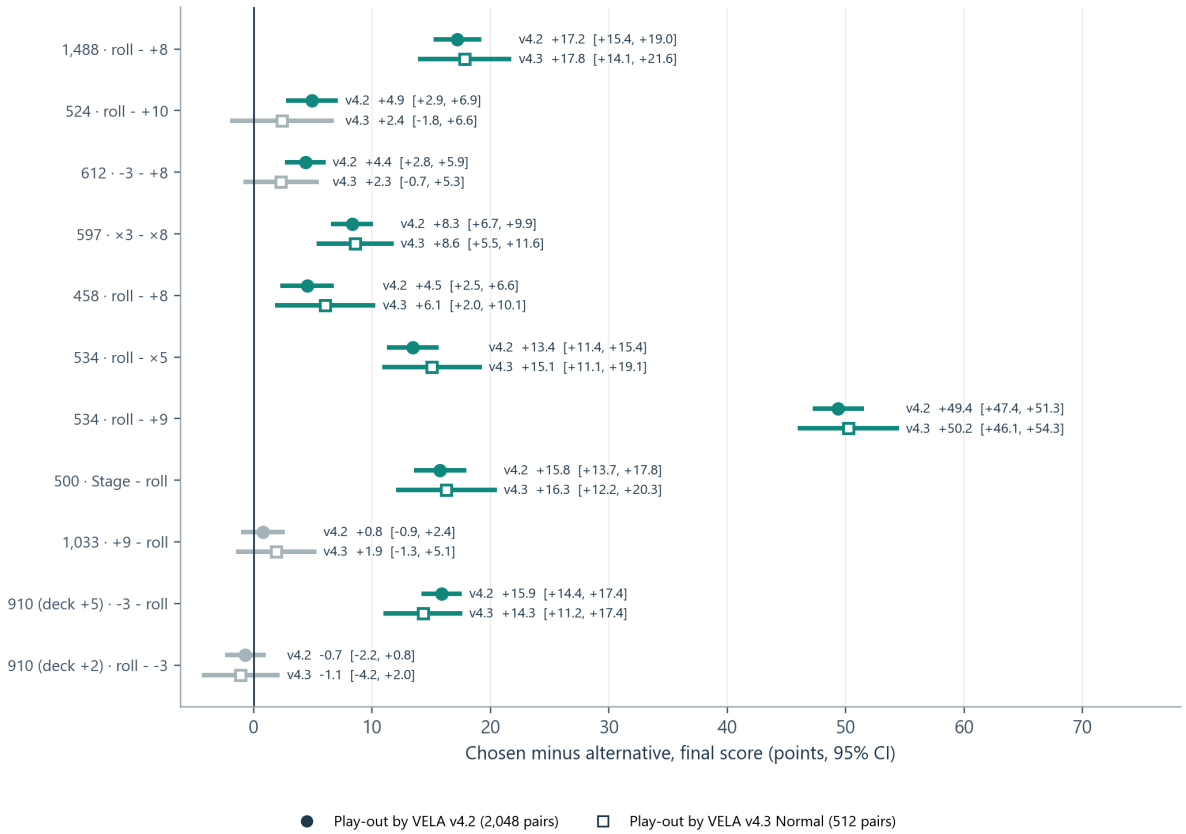


Figure 29: Play-outs that change only the first action. For each case the upper line is a play-out by VELA v4.2 and the lower line by v4.3 Normal. Green intervals lie above 0; gray ones include 0.

With VELA v4.2 playing on, the AI's choice finished ahead of the alternative in 9 of the 11 comparisons. The two that include 0 (+9 vs. roll at 1,033, and roll vs. −3 at 910 with a +2 in the deck) are positions where the AI's own values were close or tied. With v4.3 Normal playing on (512 pairs each), 7 of the 11 lay above 0. 1,033 and 910 with a +2 included 0, as they did for v4.2. 524 and 612 also included 0, but the v4.2 play-outs on the same 512 pairs gave similar results: 524 +0.47 points [-3.62, +4.56] and 612 +3.10 points [+0.06, +6.13]. With a quarter of the 2,048 pairs, the intervals are about twice as wide, and at this precision the AI that plays on makes no measurable difference. A small value gap can still mean a clear long-run difference, as at 458 (3.5 points), where both play-outs lie above 0, so the size of the value gap alone does not predict the long-run difference.

Case position	VELA v4.3	VELA v4.2	LYRA v2	Agreement
1,488 · 524 · 458 · 612 · 597	Same action	Recorded choice	Same action	Agree
534 · 536 · 500 · 538	Same action	Same action	Recorded choice	Agree
910 (deck +5)	−3	−3	−3	Agree
1,033	+9 · +11 tied	+9 · +11 tied	+9 (1.7 above +11)	Effectively agree
910 (deck +2)	−3 (0.2–0.5 apart)	−3 (0.6 apart)	Roll (0.01 apart)	Narrow split

The evidence is strongest when the value gap is large, the AIs agree, and the outcome breakdown does not hinge on a single dice result (Cases 1, 3, 6 and 8 with a +5 in the deck). When the gap is about 1 point or the AIs disagree, either choice probably costs little, and there is little reason to treat that one recommendation as the right answer. In endgame ordering, VELA v4.2 and LYRA v2 occasionally make the same mistake, as at 1,636. VELA v4.3 Normal and above correct most of these positions but also introduce some new mistakes (Sections 5.2 and 5.4).

PART A

Glossary and data

The terms used in the report and the conditions behind each calculation.

Glossary

Term	Meaning
Full hand	Five cards in hand. Landing on a draw tile gives nothing (a played card leaves the hand first, so a card play can still draw).
Deck, refill	The cards not yet drawn (30 at the start). Drawing the last card refills the deck to 30.
Stop tile (boss tile)	A tile a plain roll cannot pass. Multiplier, move and stage cards go past it.
Bonus roll	A free extra roll earned by a double on a paid roll.
Draw reachable	The action draws, or one card left in hand can draw from the landing tile.
AI value	The probability-weighted average of the AI's values for every outcome of an action; an estimate of the expected final score.
Holding value	The difference in AI value with and without a card in hand, in the same position.
Outcome breakdown	An action's value split by dice result and card drawn. The parts add back up to the value.
Play-out comparison	Two games from the same position and seed (the number that fixes the random sequence) that differ only in the first action; the score gap is measured over many seeds. VELA v4.2 or v4.3 Normal plays the rest of each game.
95% interval	The uncertainty range of a measured mean gap. If it includes 0, the sample cannot tell which action is better.
Exact value	The expected final score over every remaining possibility. Only computable in the endgame.
Search depth	How many steps ahead the AI calculates: two for VELA v4.2, one for LYRA v2. VELA v4.3 has four search strengths (Fast = v4.2, Normal, High, Max); Normal and above search three steps and up to six in close calls.

Data and conditions

Data	Details
Case positions	Positions reached in VELA v4.2 games (1,488 · 524 · 458 · 597 · 612) and LYRA v2 games (534 · 500 · 1,033 · 910). The 910 position was evaluated twice, with the last deck card set to +5 and to +2.
Outcome breakdowns	VELA v4.2: the values of the child positions in its two-step evaluation. LYRA v2: the network's values of the one-step child positions. Recombined outcome values match each AI's action value to within 0.001 points.
Holding values	Computed with the VELA v4.2 value model. The phase table uses 7,000 random positions with cards in hand from 2,000 VELA v4.2 games.
Base probabilities	Computed from the rules over all 2,659 of the 2,898 tiles where the token can stand.
Choice patterns	Decision logs of 10,000 VELA v4.2 games and 10,000 LYRA v2 games. The two sets are different games.
Play-outs	Eleven cases. VELA v4.2 plays on (2,048 pairs each) or v4.3 Normal plays on (512 pairs each, on the first 512 seeds of the v4.2 runs). Accumulating the gap between each move's exact expected transition and its actual result removes much of the dice luck (control-variate correction). Results hold for the AI that plays on; they are not guaranteed to match the gap when another AI or a person continues the game.
VELA v4.3	The production search code, run with the same model file. At Fast it chose the same action as v4.2 in all 4,312 catalog positions and gave the same scores as v4.2 in 12 sample play-out pairs.
Override experiment	VELA v4.2 played 3,000 games per rule on the same seeds, and control-variate-corrected scores were compared seed by seed. The second-best action is the one with the second-highest VELA v4.2 value.
Exact values	4,312 positions from the exact endgame catalog (0–2 rolls left). Cases 10 and 11 were solved separately.
One million games	Final scores of one million VELA v4.2 games. All 196 games at 2,600 or more and 130 lower-scoring games were replayed and matched the original records.